

# Chanyoung Park

## CONTACT INFORMATION

### Associate Professor

Data Science & Artificial Intelligence Lab (DSAIL)  
Dept. of Industrial & Systems Engineering, KAIST  
The Kim Jaechul Graduate School of AI, KAIST  
Graduate School of Data Science, KAIST

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Homepage: <https://dsail.kaist.ac.kr>

[Google Scholar](#)

## CURRENT RESEARCH

- **AI for Scientific Discovery**
  - Materials, Chemistry
  - Biology, Engineering, EDA
- **Agentic AI**
- **LLM / Multimodal Data Mining**
  - Recommender System, Graph Neural Networks
  - LLM Safety, Model Efficiency, Video Language Model
  - Scene Graph Generation, Anomaly Detection, Time Series Analysis
- **Data-Centric AI**
  - Model Robustness, Adversarial Attack/Defense
  - Imbalanced Learning, Few-shot Learning, Explainability

## POSITIONS

**KAIST**, Daejeon, South Korea

Mar 2025 – Present

- Associate Professor, Dept. of Industrial & Systems Engineering
- Associate Professor, The Kim Jaechul Graduate School of AI
- Associate Professor, Graduate School of Data Science

**KAIST**, Daejeon, South Korea

Nov 2020 – Feb 2025

- Assistant Professor, Dept. of Industrial & Systems Engineering
- Assistant Professor, The Kim Jaechul Graduate School of AI
- Assistant Professor, Graduate School of Data Science

**University of Illinois at Urbana-Champaign**, Urbana, IL, USA

Jan 2019 – Oct 2020

- Postdoctoral Reserach Fellow in Computer Science Department
  - Host: [Prof. Jiawei Han](#)
  - Topics: Multimodal data mining, Heterogeneous network representation learning

**Microsoft Research**, Beijing, China

Sep 2017 – Dec 2017

- Research Intern
  - Mentor: [Dr. Xing Xie](#)
  - Project: Collaborative Translational Metric Learning

**NAVER**, Seongnam, South Korea

Mar 2017 – Aug 2017

- Research Intern
  - Mentors: [Dr. Jung-Tae Lee](#), and [Dr. Min-Chul Yang](#)
  - Project: Click-aware Purchase Prediction with Push at the Top

## EDUCATION

**Pohang University of Science and Technology**, Pohang, South Korea

Feb 2019

- Ph.D. in Computer Science and Engineering
  - Thesis: Recommendation Framework via Matrix Factorization and Translation
  - Topics: Multimodal data mining, User behavior modeling, Recommender system

**Sogang University**, Seoul, South Korea

Feb 2014

- B.S. in Computer Science and Engineering

## AWARDS & SCHOLARSHIPS

**NVIDIA Academic Grant**

2026

- NVIDIA

**Best Paper Award**

2026

- KDD 2026 Workshop on Personal Intelligence in Agentic AI Era, Jeju, South Korea

**Best Paper Award**

2026

|                                                                                                          |           |
|----------------------------------------------------------------------------------------------------------|-----------|
| ▪ KDD 2026 Workshop on Reliable Scientific Foundation Models, Jeju, South Korea                          |           |
| <b>Google Research Award</b>                                                                             | 2026      |
| ▪ Google                                                                                                 |           |
| <b>Impact Research Award (Achievement Award)</b>                                                         | 2025      |
| ▪ KAIST                                                                                                  |           |
| <b>Best Academic Award</b>                                                                               | 2025      |
| ▪ Dept. of Industrial and Systems Engineering, KAIST                                                     |           |
| <b>Song-Am Distinguished Research Award</b>                                                              | 2024      |
| ▪ College of Engineering, KAIST                                                                          |           |
| <b>Best Academic Award</b>                                                                               | 2024      |
| ▪ Dept. of Industrial and Systems Engineering, KAIST                                                     |           |
| <b>Best Paper Award</b>                                                                                  | 2024      |
| ▪ KDD 2024 Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD), Barcelona, Spain |           |
| <b>Best Paper Award</b>                                                                                  | 2024      |
| ▪ KDD 2024 Workshop on Human-Interpretable AI, Barcelona, Spain                                          |           |
| <b>Gold Prize, 30th Samsung Humantech Paper Award</b>                                                    | 2024      |
| <b>Technology Innovation Awards</b>                                                                      | 2023      |
| ▪ College of Engineering, KAIST                                                                          |           |
| <b>Top Reviewer of NeurIPS</b>                                                                           | 2023      |
| <b>Best Paper Award</b>                                                                                  | 2023      |
| ▪ ICML 2023 Workshop on Computational Biology, Hawaii, USA                                               |           |
| <b>KAIST Excellence in Teaching Award of Year 2022</b>                                                   | 2023      |
| <b>Best Teaching Award</b>                                                                               | 2023      |
| ▪ Dept. of Industrial and Systems Engineering, KAIST                                                     |           |
| <b>ICDM Travel Award</b>                                                                                 | 2018      |
| ▪ International Conference on Data Mining, Singapore                                                     |           |
| <b>Award of excellence</b>                                                                               | 2017      |
| ▪ Microsoft Research Asia Internship Program, Beijing, China                                             |           |
| ▪ Awarded to the best intern                                                                             |           |
| <b>Naver Ph.D Fellowship</b>                                                                             | 2016      |
| ▪ Awarded to outstanding Ph.D students majoring in computer science in Korea                             |           |
| <b>RecSys Challenge 2015</b>                                                                             | 2015      |
| ▪ Ranked 10 out of 850 teams from 49 countries (Top 1.1%)                                                |           |
| ▪ Task: Click/Purchase Prediction                                                                        |           |
| <b>Dean's List</b>                                                                                       | Fall 2013 |
| ▪ Sogang University                                                                                      |           |

## PUBLICATIONS

## CONFERENCES

- [C92] Compositional Spectral Prompts for LLM-based Online Time Series Forecasting  
Seungyeon Choi, Hyunchul Kim, Jae-Gil Lee, **Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2026**)
- [C91] ELASTIC: Trajectory-Based Synchronization of Event and Tracking Data in Soccer  
Hyunsung Kim, Hoyoung Choi, Kunhee Lee, Sangwoo Seo, Tom Boomstra, Jinsung Yoon,  
**Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2026**)
- [C90] DefaultGNN: A Dual-Perspective GNN Framework for Predicting Corporate Default from  
Buyer-Seller Transaction Networks  
Junghoon Kim, Hyunsung Kim, Seungyeon Choi, Kyoungyong Park, Jihun Lee, Yonggu Ji,  
**Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2026**)  
(**Applied Research Track**)
- [C89] A Behavioral Trait Leaks into Preferences: Diagnosing Trait Interference in LLM User Simulators  
Chaehyun Kim, Sein Kim, Hongseok Kang, **Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2026**) (**Short Paper**)

- [C88] HUGETIME at eRisk 2026: Bucket-wise Evidence Pruning for Contextualized Early Depression Detection  
Ilho Shin, Hyunchul Kim, **Chanyoung Park**  
The Conference and Labs of the Evaluation Forum (**CLEF 2026**)  
*Ranked 1st in Task 2: Contextualized Early Detection of Depression at CLEF eRisk 2026*
- [C87] Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs  
Kibum Kim, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, **Chanyoung Park**  
European Conference on Computer Vision (**ECCV 2026**)
- [C86] Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding  
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, **Chanyoung Park**  
European Conference on Computer Vision (**ECCV 2026**)
- [C85] PathCRF: Ball-Free Soccer Event Detection via Possession Path Inference from Player Trajectories  
Hyunsung Kim, Kunhee Lee, Sangwoo Seo, Sang-Ki Ko, Jinsung Yoon, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026**)
- [C84] Learning Probabilistic Compositional Representation of Crystalline Materials  
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Jimeng Sun, Tianfan Fu, Marinka Zitnik, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026**) (**AI for Science Track**)
- [C83] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs  
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**  
International Conference on Machine Learning (**ICML 2026**)
- [C82] Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented Agents  
Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, **Chanyoung Park**  
International Conference on Machine Learning (**ICML 2026**)
- [C81] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction  
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn  
International Conference on Machine Learning (**ICML 2026**)
- [C80] Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for Time-Series Foundation Models  
Jihye Na, Patara Trirat, **Chanyoung Park**, Jae-Gil Lee  
International Conference on Machine Learning (**ICML 2026**)
- [C79] SelfJudge: Faster Speculative Decoding via Self-Supervised Judge Verification  
Kanghoon Yoon, Minsub Kim, Sungjae Lee, Joonhyung Lee, Sunghyeon Woo, Yeonjun In, Se Jung Kwon, **Chanyoung Park**, Dongsoo Lee  
International Conference on Machine Learning (**ICML 2026**)
- [C78] Reasoning Structure Matters for Safety Alignment of Reasoning Models  
Yeonjun In, Wonjoong Kim, Sangwu Park, **Chanyoung Park**  
The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**)
- [C77] IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra  
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, Kibum Kim, **Chanyoung Park**  
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C76] CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs  
Jiwan Kim, Kibum Kim, Sangwoo Seo, **Chanyoung Park**  
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C75] Token-Efficient Item Representation via Images for LLM Recommender Systems  
Kibum Kim, Sein Kim, HongSeok Kang, Jiwan Kim, Heewoong Noh, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Julian McAuley, **Chanyoung Park**  
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)

- [C74] PULSE: Socially-Aware User Representation Modeling Toward Parameter-Efficient Graph Collaborative Filtering  
Doyun Choi\*, Cheonwoo Lee\*, Biniyam Aschalew Tolera, Taewook Ham, **Chanyoung Park**, Jaemin Yoo  
The Web Conference (**WWW 2026**)
- [C73] RAG-Enhanced Collaborative LLM Agents for Drug Discovery  
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, Tommaso Biancalani, **Chanyoung Park**, Gabriele Scalia  
The 40th Annual AAAI Conference on Artificial Intelligence (**AAAI 2026**)
- [C72] 3D Interaction Geometric Pre-training for Molecular Relational Learning  
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Minkai Xu, Hanchen Wang, Tianfan Fu, **Chanyoung Park**  
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025 (Spotlight)**)
- [C71] Disentangling Hyperedges through the Lens of Category Theory  
Yoonho Lee, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, **Chanyoung Park**  
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)
- [C70] Training Robust Graph Neural Networks by Modeling Noise Dependencies  
Yeonjun In, Kanghoon Yoon, Sukwon Yun, Kibum Kim, Sungchul Kim, **Chanyoung Park**  
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)
- [C69] Is Safety Standard Same for Everyone? User-Specific Safety Evaluation of Large Language Models  
Yeonjun In, Wonjoong Kim, Kanghoon Yoon, Sungchul Kim, Mehrab Tanjim, Kibum Kim, **Chanyoung Park**  
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025 Findings**)
- [C68] Disambiguation in Conversational Question Answering in the Era of LLM: A Survey  
Md Mehrab Tanjim\*, Yeonjun In\*, Xiang Chen, Victor S. Burszty, Ryan A. Rossi, Sungchul Kim, Guang-Jie Ren, Vaishnavi Muppala, Shun Jiang, Yongsung Kim, **Chanyoung Park**  
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025**)
- [C67] Target Circuit Matching in Large-Scale Netlists using GNN-Based Region Prediction  
Sangwoo Seo, Jimin Seo, Yoonho Lee, Donghyeon Kim, Hyejin Shin, Banghyun Sung, **Chanyoung Park**  
International Conference on Computer-Aided Design (**ICCAD 2025**)
- [C66] Lost in Sequence: Do Large Language Models Understand Sequential Recommendation?  
Sein Kim\*, HongSeok Kang\*, Kibum Kim, Jiwan Kim, Donghyun Kim, Min-Chul Yang, Kwangjin Oh, Julian McAuley, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C65] Oldie but Goodie: Re-illuminating Label Propagation on Graphs with Partially Observed Features  
Sukwon Yun, Xin Liu, Yunhak Oh, Junseok Lee, Tianlong Chen, Tsuyoshi Murata, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C64] Electron-Informed Coarse-Graining Molecular Representation Learning for Real-World Molecular Physics  
Gyoung S. Na, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C63] On Measuring Unnoticeability of Graph Adversarial Attacks: Observations, New Measure, and Applications  
Hyeonsoo Jo\*, Hyunjin Hwang\*, Fanchen Bu, Soo Yong Lee, **Chanyoung Park**, Kijung Shin  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C62] Thickness-aware E(3)-Equivariant 3D Mesh Neural Networks  
Sungwon Kim, Namkyeong Lee, Yunyoung Doh, Seungmin Shin, Guimok Cho, Seung-Won Jeon, Sangkook Kim, **Chanyoung Park**  
International Conference on Machine Learning (**ICML 2025**)

- [C61] Global Context-aware Representation Learning for Spatially Resolved Transcriptomics  
Yunhak Oh\*, Junseok Lee\*, Yeongmin Kim, Sangwoo Seo, Namkyeong Lee, **Chanyoung Park**  
International Conference on Machine Learning (**ICML 2025**)
- [C60] Disentangling and Generating Modalities for Recommendation in Missing Modality Scenarios  
Jiwan Kim, Hongseok Kang, Sein Kim, Kibum Kim, **Chanyoung Park**  
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)
- [C59] Dynamic Time-aware Continual User Representation Learning  
Seungyoon Choi, Sein Kim, Hongseok Kang, Wonjoong Kim, **Chanyoung Park**  
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)
- [C58] Subgraph Federated Learning for Local Generalization  
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**  
The Thirteenth International Conference on Learning Representations (**ICLR 2025 (Oral)**)
- [C57] Weakly Supervised Video Scene Graph Generation via Natural Language Supervision  
Kibum Kim, Kanghoon Yoon, Yeonjun In, Jaehyeong Jeon, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**  
The Thirteenth International Conference on Learning Representations (**ICLR 2025**)
- [C56] Self-Supervised Diffusion Models for Electron-Aware Molecular Representation Learning  
Gyoung S. Na, **Chanyoung Park**  
The Thirteenth International Conference on Learning Representations (**ICLR 2025**)
- [C55] Diversify-verify-adapt: Efficient and Robust Retrieval-Augmented Ambiguous Question Answering  
Yeonjun In, Sungchul Kim, Ryan A. Rossi, Mehrab Tanjim, Tong Yu, Ritwik Sinha, **Chanyoung Park**  
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025**)
- [C54] SIMPLOT: Enhancing Chart Question Answering by Distilling Essentials  
Wonjoong Kim\*, Sangwu Park\*, Yeonjun In, Seokwon Han, **Chanyoung Park**  
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025 Findings**)
- [C53] RA-SGG: Retrieval-Augmented Scene Graph Generation Framework via Multi-Prototype Learning  
Kanghoon Yoon, Kibum Kim, Jaehyeong Jeon, Yeonjun In, Donghyun Kim, **Chanyoung Park**  
Thirty-Ninth AAAI Conference on Artificial Intelligence (**AAAI 2025**)
- [C52] Revisiting Fake News Detection: Towards Temporality-aware Evaluation by Leveraging Engagement Earliness  
Junghoon Kim\*, Junmo Lee\*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**  
The 18th ACM International Conference on Web Search and Data Mining (**WSDM 2025 (Oral)**)
- [C51] Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge  
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, **Chanyoung Park**  
Thirty-eighth Conference on Neural Information Processing Systems (**NeurIPS 2024**)
- [C50] Debaised Graph Poisoning Attack via Contrastive Surrogate Objective  
Kanghoon Yoon, Yeonjun In, Namkyeong Lee, Kibum Kim, **Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2024**)
- [C49] Vision Language Model is NOT All You Need: Augmentation Strategies for Molecule Language Models  
Namkyeong Lee, Siddhartha Laghuvarapu, **Chanyoung Park**, Jimeng Sun  
ACM International Conference on Information and Knowledge Management (**CIKM 2024**)
- [C48] Semantic Diversity-aware Prototype-based Learning for Unbiased Scene Graph Generation  
Jaehyeong Jeon, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**  
European Conference on Computer Vision (**ECCV 2024**)
- [C47] Multiplexed Immunofluorescence Image Analysis through an Efficient Multiplex Network  
Sukwon Yun, Jie Peng, Alexandro Trevino, **Chanyoung Park**, Tianlong Chen  
European Conference on Computer Vision (**ECCV 2024**)

- [C46] Large Language Models meet Collaborative Filtering: An Efficient All-round LLM-based Recommender System  
Sein Kim\*, Hongseok Kang\*, Seungyeon Choi, Donghyun Kim, Minchul Yang, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)
- [C45] Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck  
Sangwoo Seo, Sungwon Kim, Jihyeong Jung, Yoonho Lee, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)
- [C44] Unsupervised Episode Generation for Graph Meta-learning  
Jihyeong Jung, Sangwoo Seo, Sungwon Kim, **Chanyoung Park**  
International Conference on Machine Learning (**ICML 2024**)
- [C43] LLM4SGG: Large Language Models for Weakly Supervised Scene Graph Generation  
Kibum Kim, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**  
IEEE/CVF Computer Vision and Pattern Recognition Conference (**CVPR 2024**)
- [C42] Self-guided Robust Graph Structure Refinement  
Yeonjun In, Kanghoon Yoon, Kibum Kim, Kijung Shin, **Chanyoung Park**  
The Web Conference 2024 (**WWW 2024 (Oral)**)
- [C41] DSLR: Diversity Enhancement and Structure Learning for Rehearsal-based Graph Continual Learning  
Seungyeon Choi\*, Wonjoong Kim\*, Sungwon Kim, Yeonjun In, Sein Kim, **Chanyoung Park**  
The Web Conference 2024 (**WWW 2024 (Oral)**)
- [C40] Adaptive Self-training Framework for Fine-grained Scene Graph Generation  
Kibum Kim\*, Kanghoon Yoon\*, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**  
Twelfth International Conference on Learning Representations (**ICLR 2024**)
- [C39] Sterling: Synergistic Representation Learning on Bipartite Graphs  
Baoyu Jing, Yuchen Yan, Kaize Ding, **Chanyoung Park**, Yada Zhu, Huan Liu, Hanghang Tong  
Thirty-Eighth AAAI Conference on Artificial Intelligence (**AAAI 2024 (Oral)**)
- [C38] Interpretable Prototype-based Graph Information Bottleneck  
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**  
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**) (**Gold Prize, 30th Samsung Humantech Paper Award**)
- [C37] Density of States Prediction of Crystalline Materials via Prompt-guided Multi-Modal Transformer  
Namkyeong Lee\*, Heewoong Noh\*, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park**  
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**)
- [C36] MUSE: Music Recommender System with Shuffle Play Recommendation Enhancement  
Yunhak Oh\*, Sukwon Yun\*, Dongmin Hyun, Sein Kim, **Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2023**)
- [C35] Class Label-aware Graph Anomaly Detection  
Junghoon Kim, Yeonjun In, Kanghoon Yoon, Junmo Lee, **Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2023**) (Short Paper)
- [C34] Structural Mixup for Graph Neural Networks  
Junghun Kim\*, Sukwon Yun\*, **Chanyoung Park**  
ACM International Conference on Information and Knowledge Management (**CIKM 2023**) (Short Paper)
- [C33] Similarity Preserving Adversarial Graph Contrastive Learning  
Yeonjun In\*, Kanghoon Yoon\*, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C32] Shift-Robust Molecular Relational Learning with Causal Substructure  
Namkyeong Lee, Kanghoon Yoon, Gyoung S. Na, Sein Kim, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)



- [C31] Task-Equivariant Graph Few-shot Learning  
Sungwon Kim, Junseok Lee, Namkyeong Lee, Wonjoong Kim, Seungyeon Choi, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C30] Task Relation-aware Continual User Representation Learning  
Sein Kim, Namkyeong Lee, Donghyun Kim, Min-Chul Yang, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C29] Conditional Graph Information Bottleneck for Molecular Relational Learning  
Namkyeong Lee, Dongmin Hyun, Gyoung S. Na, Sungwon Kim, Junseok Lee, **Chanyoung Park**  
International Conference on Machine Learning (**ICML 2023**)
- [C28] MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation  
Kibum Kim, Dongmin Hyun, Sukwon Yun, **Chanyoung Park**  
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2023**)
- [C27] Unbiased Heterogeneous Scene Graph Generation with Relation-aware Message Passing Neural Network  
Kanghoon Yoon\*, Kibum Kim\*, Jinyoung Moon, **Chanyoung Park**  
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023**)
- [C26] Heterogeneous Graph Learning for Multi-modal Medical Data Analysis  
Sein Kim, Namkyeong Lee, Junseok Lee, Dongmin Hyun, **Chanyoung Park**  
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023 (Oral)**)
- [C25] Generating Multiple-Length Summaries via Reinforcement Learning for Unsupervised Sentence Summarization  
Dongmin Hyun, Xiting Wang, **Chanyoung Park**, Xing Xie, Hwanjo Yu  
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2022 Findings**)
- [C24] Set2Box: Similarity Preserving Representation Learning for Sets  
Geon Lee, **Chanyoung Park**, Kijung Shin  
The IEEE International Conference on Data Mining (**ICDM 2022**)
- [C23] LTE4G: Long-Tail Experts for Graph Neural Networks  
Sukwon Yun, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**  
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C22] Relational Self-Supervised Learning on Graphs  
Namkyeong Lee, Dongmin Hyun, Junseok Lee, **Chanyoung Park**  
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C21] Beyond Learning from Next Item: Sequential Recommendation via Personalized Interest Sustainability  
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu  
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C20] Nonlinearity Encoding for Extrapolation of Neural Networks  
Gyoung S. Na, **Chanyoung Park**  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2022**)
- [C19] GraFN: Semi-Supervised Node Classification on Graph with Few Labels via Non-Parametric Distribution Assignment  
Junseok Lee, Yunhak Oh, Yeonjun In, Namkyeong Lee, Dongmin Hyun, **Chanyoung Park**  
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)
- [C18] AHP: Learning to Negative Sample for Hyperedge Prediction  
Hyunjin Hwang, Seungwoo Lee, **Chanyoung Park**, Kijung Shin  
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)
- [C17] Augmentation-Free Self-Supervised Learning on Graphs  
Namkyeong Lee, Junseok Lee, **Chanyoung Park**  
Thirty-Sixth AAAI Conference on Artificial Intelligence (**AAAI 2022**)

- [C16] Learnable Structural Semantic Readout for Graph Classification  
Dongha Lee, Su Kim, Seonghyeon Lee, **Chanyoung Park**, Hwanjo Yu  
IEEE International Conference on Data Mining (**ICDM 2021**) (Short Paper)
- [C15] Bootstrapping User and Item Representations for One-Class Collaborative Filtering  
Dongha Lee, Seongku Kang, Hyunjun Ju, **Chanyoung Park**, Hwanjo Yu  
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2021**)
- [C14] HDMI: High-order Deep Multiplex Infomax  
Baoyu Jing, **Chanyoung Park**, Hanghang Tong  
The Web Conference (**WWW 2021**)
- [C13] Interest Sustainability-Aware Recommender System  
Dongmin Hyun, Junsu Cho, **Chanyoung Park**, Hwanjo Yu  
IEEE International Conference on Data Mining (**ICDM 2020**)
- [C12] Unsupervised Differentiable Multi-aspect Network Embedding  
**Chanyoung Park**, Carl Yang, Qi Zhu, Donghyun Kim, Hwanjo Yu and Jiawei Han  
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2020**)
- [C11] Unsupervised Attributed Multiplex Network Embedding  
**Chanyoung Park**, Donghyun Kim, Jiawei Han, Hwanjo Yu  
Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**)
- [C10] Task-Guided Pair Embedding in Heterogeneous Network  
**Chanyoung Park**, Donghyun Kim, Qi Zhu, Jiawei Han, Hwanjo Yu  
ACM International Conference on Information and Knowledge (**CIKM 2019**)
- [C9] BHIN2vec: Balancing the Type of Relation in Heterogeneous Information Network  
Seonghyun Lee, **Chanyoung Park**, Hwanjo Yu  
ACM International Conference on Information and Knowledge (**CIKM 2019**)
- [C8] Action Space Learning for Heterogeneous User Behavior Prediction  
Dongha Lee, **Chanyoung Park**, Hyunjun Ju, Junyoung Hwang, Hwanjo Yu  
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)
- [C7] Sequential and Diverse Recommendation with Long Tail  
Yejin Kim, Kwangseob Kim, **Chanyoung Park**, Hwanjo Yu  
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)
- [C6] Collaborative Translational Metric Learning  
**Chanyoung Park**, Donghyun Kim, Xing Xie, Hwanjo Yu  
IEEE International Conference on Data Mining (**ICDM 2018**)
- [C5] Review Sentiment-Guided Scalable Deep Recommender System  
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu  
ACM SIGIR conference on Research and Development in Information Retrieval (**SIGIR 2018**)  
(Short Paper)
- [C4] Do “Also-Viewed” Products Help User Rating Prediction?  
**Chanyoung Park**, Donghyun Kim, Jinoh Oh, Hwanjo Yu  
International World-Wide Web Conference (**WWW 2017**)
- [C3] Convolutional Matrix Factorization for Document Context-Aware Recommendation  
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu  
ACM international conference on Recommender System (**RecSys 2016**)
- [C2] TRecSo: Enhancing Top-k Recommendation With Social Information  
**Chanyoung Park**, Donghyun Kim, Jinoh Oh, Hwanjo Yu  
International World-Wide Web Conference (**WWW 2016**) (Poster)
- [C1] Predicting User Purchase in E-commerce by Comprehensive Feature Engineering and Decision  
Boundary Focused Under-Sampling  
**Chanyoung Park**, Donghyun Kim, Jinoh Oh, Hwanjo Yu  
ACM international conference on Recommender System Challenge (**ACM RecSysChallenge 2015**,  
10th place, top 1.1%)

## JOURNALS



- [J14] A New DAWN for Fake News Detection: Leveraging Engagement Earliness for Temporality-aware Evaluation  
Junghoon Kim\*, Junmo Lee\*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**  
**ACM Transactions on Intelligent Systems and Technology (2026)**
- [J13] Physics-Embedded Graph Neural Operator for Interaction-Controlled Colloidal Aggregation  
Yongjoon Choe\*, Sungwon Kim\*, Susan E. Burns, **Chanyoung Park**  
**Water Research (2026)**
- [J12] Single-cell RNA Sequencing Data Imputation Using Bi-level Feature Propagation  
Junseok Lee\*, Sukwon Yun\*, Yeongmin Kim, Tianlong Chen, Manolis Kellis, **Chanyoung Park**  
**Briefings in Bioinformatics (2024)**
- [J11] Deep Single-cell RNA-seq Data Clustering with Graph Prototypical Contrastive Learning  
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**  
**Bioinformatics (2023)**
- [J10] Self-Supervised Graph Representation Learning via Positive Mining  
Namkyeong Lee, Junseok Lee, **Chanyoung Park**  
**Information Sciences (2022.9)**
- [J9] Learning to Utilize Auxiliary Reviews for Recommendation  
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu  
**Information Sciences (2021.2)**
- [J8] Deep Multiplex Graph Infomax: Attentive Multiplex Network Embedding using Global Information  
**Chanyoung Park**, Jiawei Han, Hwanjo Yu  
**Knowledge-Based Systems (2020.6)**
- [J7] Improved prediction of miRNA–disease associations based on matrix completion with network regularization  
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park  
**Cells (2020.4) (SCIE)**
- [J6] Click-aware Purchase Prediction with Push at the Top  
**Chanyoung Park**, Donghyun Kim, Min-Chul Yang, Jung-Tae Lee, Hwanjo Yu  
**Information Sciences (2020.6)**
- [J5] Inferring miRNA-disease Interactions using Probabilistic Matrix Factorization  
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park  
**Journal of Biomedical Informatics (2020.2)**
- [J4] An Encoder-Decoder Switch Network for Purchase Prediction  
**Chanyoung Park**, Donghyun Kim, Hwanjo Yu  
**Knowledge-Based Systems (2019.12)**
- [J3] Target-aware Convolutional Neural Network for Target-level Sentiment Analysis  
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu  
**Information Sciences (2019.7)**
- [J2] Deep Hybrid Recommender Systems via Exploiting Document Context and Statistics of Items  
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu  
**Information Sciences (2017.11)**
- [J1] Improving top-K recommendation with truster and trustee relationship in user trust network  
**Chanyoung Park**, Donghyun Kim, Jinoh Oh, Hwanjo Yu  
**Information Sciences (2016.12)**

#### WORKSHOPS

- [W28] Tokenizing Antibody–Antigen Complex Structures via Energy-Based Surrogates  
Junseok Lee, Yunhak Oh, Namkyeong Lee, Hyunchul Kim, **Chanyoung Park**  
Workshop on Reliable Scientific Foundation Models (**KDD 2026 Workshop**)
- [W27] Bridging Circuit Graphs and Language: A Cross-Modal Transformer for Netlist-to-Text Alignment and Generation  
Sangwoo Seo, Hyunsung Kim, Hyunchul Kim, Byeonggon Kang, Rex Ying, **Chanyoung Park**  
Workshop on Frontiers in Graph Machine Learning for the Large Model Era (**KDD 2026 Workshop**)

- [W26] Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations  
Sangwoo Seo, Hyunsung Kim, Jiwan Kim, **Chanyoung Park**  
Workshop on Reliable Scientific Foundation Models (**KDD 2026 Workshop**)
- [W25] A Structured LLM Framework for Inorganic Material Synthesis Planning  
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**  
Workshop on Agentic AI for Scientific and Societal Advances (**KDD 2026 Workshop**)
- [W24] Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented Agents  
Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, **Chanyoung Park**  
Workshop on Evaluation and Trustworthiness of Agentic AI (**KDD 2026 Workshop**)
- [W23] Compositional Spectral Prompts for LLM-based Online Time Series Forecasting  
Seungyeon Choi, Hyunchul Kim, Jae-Gil Lee, **Chanyoung Park**  
Workshop on Mining and Learning from Time Series (**KDD 2026 Workshop**)
- [W22] Personalize-then-Store: Benchmarking and Learning Personalized Memory for Long-Horizon Agents  
Yeonjun In, Wonjoong Kim, Sangwu Park, Kanghoon Yoon, **Chanyoung Park**  
Workshop on Personal Intelligence in Agentic AI Era (**KDD 2026 Workshop**)
- [W21] Rethinking Failure Attribution in Multi-Agent Systems: A Multi-Perspective Benchmark and Evaluation  
Yeonjun In, Mehrab Tanjim, Jayakumar Subramanian, Sungchul Kim, Uttaran Bhattacharya, Wonjoong Kim, Sangwu Park, Somdeb Sarkhel, **Chanyoung Park**  
Workshop on Agentic Software Engineering (**KDD 2026 Workshop**)
- [W20] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback  
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon, Hyunsik Jeon, **Chanyoung Park**  
Workshop on Agentic Software Engineering (**KDD 2026 Workshop**)
- [W19] Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding  
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, **Chanyoung Park**  
The Fourth International Workshop on Resource-Efficient Learning for Knowledge Discovery (**KDD 2026 Workshop**)
- [W18] Revisiting CF-Integrated LLM Recommenders through Dimensional Collapse  
Jimin Seo, HongSeok Kang, Sein Kim, Jiwan Kim, **Chanyoung Park**  
Geometric Space, Architecture and Learning Objectives for Large Pre-Trained Models (**KDD 2026 Workshop**)
- [W17] Physics-Embedded Graph Neural Operator for Interaction-Controlled Colloidal Aggregation  
Yongjoon Choe, Sungwon Kim, Susan E. Burns, **Chanyoung Park**  
Workshop on Reliable Scientific Foundation Models (**KDD 2026 Workshop**)
- [W16] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs  
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**  
**ICLR 2026 Workshop on AI and Partial Differential Equations (Oral) (ICLR 2026 Workshop)**
- [W15] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback  
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**  
**ICLR 2026 Workshop on AI with Recursive Self-Improvement (RSI 2026) (ICLR 2026 Workshop)**
- [W14] MSP-LLM: A Unified Large Language Model Framework for Complete Material Synthesis Planning  
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**  
**AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)**
- [W13] Synergistic Multi-Task Learning for Electronic Density of States Prediction  
Kunmin Jang, Dongik Park, Jaewon Bae, **Chanyoung Park**  
**AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)**

- [W12] Learning k-Resolved Electronic Structure via Soft Energy Occupancy Prediction  
Dongik Park, Kunmin Jang, Jaewon Bae, Dongin Kim, **Chanyoung Park**  
**AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)**
- [W11] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction  
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn  
**ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (ICLR 2026 Workshop)**
- [W10] RAG-Enhanced Collaborative LLM Agents for Drug Discovery  
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, **Chanyoung Park**, Gabriele Scalia  
**MLGenX Workshop at ICLR 2025 (MLGenX 2025 Spotlight)**
- [W9] 3D Interaction Geometric Pre-training for Molecular Relational Learning  
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Tianfan Fu, **Chanyoung Park**  
**NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX)**
- [W8] Subgraph Federated Learning for Local Generalization  
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**  
**KDD 2024 Workshop on Human-Interpretable AI (Oral) (Best Paper Award)**
- [W7] Interpretable Graph Model with Prototype-Based Graph Information Bottleneck  
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**  
**KDD 2024 Workshop on Human-Interpretable AI (Oral) (Best Paper Award)**
- [W6] Stoichiometry Representation Learning with Polymorphic Crystal Structures  
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Tianfan Fu, Jimeng Sun, **Chanyoung Park**  
**NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)**
- [W5] Electron-Derived Molecular Representation Learning for Real-World Molecular Physics  
Gyoung S. Na, **Chanyoung Park**  
**NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)**
- [W4] Deep single-cell RNA-seq data clustering with graph prototypical contrastive learning  
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**  
**ICML 2023 Workshop on Computational Biology**
- [W3] Denoising single-cell RNA-seq data using Feature Propagation  
Sukwon Yun, Junseok Lee, **Chanyoung Park**  
**ICML 2023 Workshop on Computational Biology (Oral) (Best Paper Award)**
- [W2] Predicting Density of States via Multi-modal Transformer  
Namkyeong Lee, Heewoong Noh, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park**  
**ICLR 2023 Workshop on Machine Learning for Materials (ML4Materials)**
- [W1] Shift-Robust Node Classification via Graph Adversarial Clustering  
Qi Zhu, Chao Zhang, **Chanyoung Park**, Carl Yang, Jiawei Han  
**NeurIPS 2022 GLFrontiers Workshop**

#### PATENT

- [P1] Method for context-aware recommendation by considering contextual information of document and apparatus for the same  
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu  
No. 10-18771610000, issued July 2018.

#### PREPRINT

- [A1] Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations  
Sangwoo Seo, Hyunsung Kim, Jiwan Kim, **Chanyoung Park**

#### TEACHING

##### Instructor

- IE343: Statistical Machine Learning, KAIST

2021S, 2022S, 2023S, 2024S, 2026S

- DS535 (KSE801): Recommender System and Machine Learning on Graphs, KAIST 2021F, 2022F, 2023F, 2024F
- CoE202: Basics of Artificial Intelligence, KAIST 2021F
- DS503: Machine Learning for Data Science (Deep Learning), KAIST 2022S, 2023S, 2024S

#### **Teaching Assistant**

- CSED352: Data Communications, POSTECH 2014F
- CSED312: Operating System, POSTECH 2016F

### **PROFESSIONAL SERVICES**

#### **AREA CHAIR / SENIOR PROGRAM COMMITTEE**

- The International Conference on Machine Learning (ICML) [2025-present]
- The International Conference on Learning Representations (ICLR) [2025-present]
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) [2025-present]
- Conference on Neural Information Processing Systems (NeurIPS) [2024-present]
- AAAI Conference on Artificial Intelligence (AAAI) [2026-present]
- ICML 2024 AI4Science Workshop [2024-present]
- NeurIPS Workshop (AI for Science: from Theory to Practice) [2023-present]

#### **EVENT ORGANIZATIONS**

- Organizer, KDD'26 Workshop on SciSoc Agents & LLMs [2026]
- General Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2025]
- Proceedings Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2023]

#### **JOURNAL EDITORSHIP**

- Action Editor in Transactions on Machine Learning Research ([TMLR](#))
- Frontiers in Artificial Intelligence ([Graph representation learning: Methods and Applications](#))
- Frontiers in Big Data ([Reviews in Big Data Networks: 2022](#))